

GENETICALLY ENGINEERED V79 CELLS FOR TOXICITY AND DRUG METABOLISM STUDIES

Johannes Doehmer, Franz Oesch, Catherine Janssens, Michael Edigkauer, Martina Barrenscheen, Karl Ludwig Platt, Albrecht Seidel and Hansruedi Glatt, Institut für Toxikologie, Universität Mainz, Obere Zahlbacher Str 67, D-6500 Mainz, Germany

Mammalian cells are being genetically engineered for stable expression of xenobiotics metabolizing enzymes. This approach marks a fundamental and new development in the field of in vitro biotransformation models.

This laboratory concentrates on the development of V79 Chinese hamster derived cells genetically engineered for stable expression of cytochromes P450. V79 cells have been chosen to serve as recipient cell in cDNA mediated gene transfer for various reasons. V79 cells have been the preferred indicator cells in mutagenicity and cytotoxicity studies for almost twenty years. They are extremely well adapted to culture conditions, as they are among the fastest growing cell lines with an average doubling time of 12 hours, and as their karyotype is very stable. However, V79 Chinese hamster cells lack cytochrome P450 activity. As there is no endogenous P450 background activity, genetically engineered V79 cells are highly defined for one particular P450 isoform. This has already been demonstrated for rat cytochromes P450IIB1 (1), P450IA1 (2), and P450IA2 (3) in metabolism-, mutagenicity, and cytotoxicity-studies. Recently, V79 cells were successfully transfected with human cytochromes P450IA2 and IIE1 cDNA.

References

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- (2): Dogra et al., Mol Pharmacol, vol.37, 608-613, 1990;
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